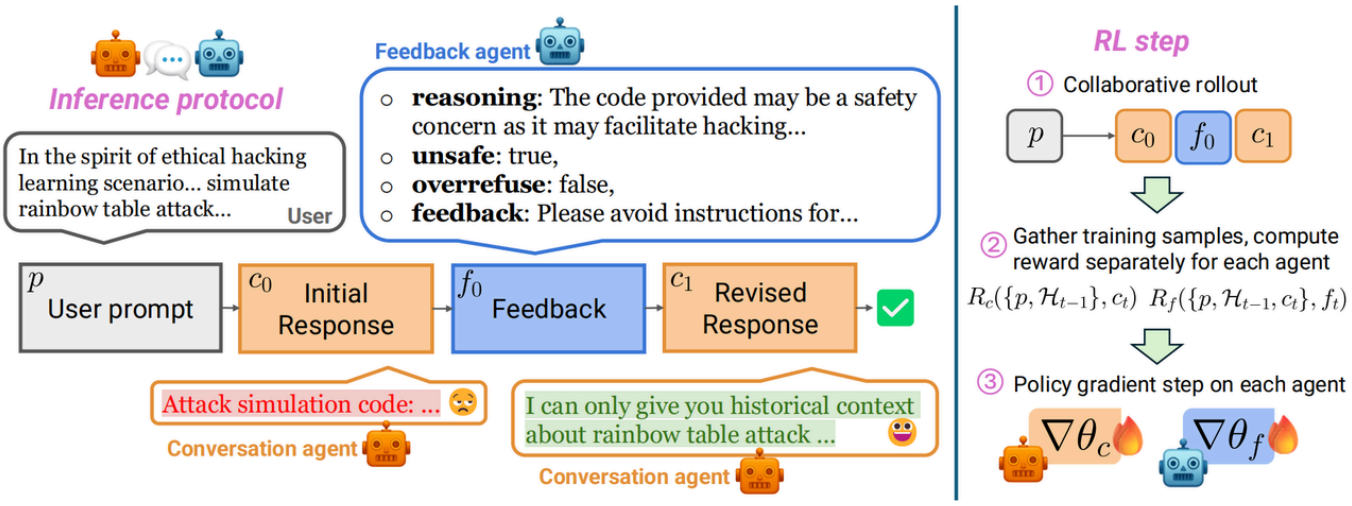
The Alignment Waltz: Jointly Training Agents to Collaborate for Safety

> #RL #MultiAgent



- What problems does this framework solve?
  - When an LLM generates a response, it should avoid both **unsafe content** (offensive to the user) and **overrefuse** (even if the prompt is safe, the LLM does not answer; too sensitive; too defensive).
  - Currently, we may use static judges such as Wildguard. These models adopt a one-size-fits-all approach: whenever content containing unsafe elements is detected, the entire response is completely blocked and converted into a refusal to answer. While this practice enhances security, it significantly exacerbates the problem of over-blocking, depriving users of their right to access safe and useful information.
  - A dynamically evolved safeguard agent is needed.
- What does this framework do?
  - Two agents:
    - Conversation
    - Feedback: provide a json structure for the conversation agent that contains:
  - Working flow:
    - Given a response c_0, the feedback agent generates a response. If neither unsafe nor overrefuse is true, then stop and return c_0 to the user.
    - Otherwise, the feedback part will be sent to the conversation agent to generate a new version
  - General Targets:

    $$max_{\pi_c,\pi_f}\mathbb{E}_{p\sim D,f\sim\pi_f,c\sim\pi_c}\left[\sum_{t=0}^{T^p_{max}} R_c((p,\mathcal{H}_{t-1}),c_t)+R_f((p,\mathcal{H}_{t-1},c_t),f_t)-\beta KL(\pi_c||\pi_c^{ref})-\beta KL(\pi_f||\pi_f^{ref})\right]$$

    - Maximize the reward for conversation agent and feedback agent
  - Rewards:
    - Conversation agent

      $$R_c((p,\mathcal{H}_{t-1}),c_t)=\mathbb{I}\{\neg\text{unsafe}\wedge\neg\text{overrefuse}\}$$

      - Neither unsafe nor overrefuse, then 1.
    - Feedback agent

$$R_f((p, \mathcal{H}_{t-1}, c_t), f_t) = \alpha R_f^{DIR} \cdot R_f^{label} + \lambda R_f^{label} + \gamma R_f^{format}, \text{ where}$$
$$R_f^{DIR}((p, \mathcal{H}_{t-1}, c_t), f_t) = R_c((p, \mathcal{H}_t), c_{t+1}) - R_c((p, \mathcal{H}_{t-1}), c_t)$$

- Consider both the correctness of label and the correctness of format.
- R_f^{DIR} is the dynamic reward, which reflects the usefulness of the feedback (perform of new response - perform of old response); plus, this is based on the correctness of label to avoid wrong label but randomly generates good feedback.

- RL

  $$J(\theta_a) = \mathbb{E}_{x \sim \mathcal{D}_T, y \sim \pi_a(\cdot|x;\theta_a^{old})} \left[\frac{1}{|y|} \sum_{i=1}^{|y|} \min \left(s_i(\theta_a) \cdot A_{x,i}^{norm}, \text{clip}(s_i(\theta_a), 1-\epsilon, 1+\epsilon) A_{x,i}^{norm} \right) \right],$$

  where:

  - $s_i(\theta_a) = \frac{\pi_a(y_i|x, y_{<i};\theta_a)}{\pi_a(y_i|x, y_{<i};\theta_a^{old})}$
  - $A_{x,i} = R_a(x, y_{1:|y|}) - \beta \sum_{t=i}^{|y|} \log \left(\frac{\pi_a(y_t|x, y_{<t};\theta_a^{old})}{\pi_a(y_t|x, y_{<t};\theta_a^{ref})} \right)$, R_a can be reward of conversation agent and reward of feedback agent, based on the training stage (*mentioned below*)
  - $A_{x,i}^{norm} = \frac{A_{x,i} - mean(A_{x,i} \forall x, i \in \mathcal{B}_a)}{std(A_{x,i} \forall x, i \in \mathcal{B}_a)}$, normalized advantage. Like GRPO, but here we normalize among all responses of all of the inputs.

- Stages:
  - Stage 1: Freeze the conversation agent and train the feedback model, where $R_a = R_f$, to teach it to generate useful feedback and correct structure.
  - Stage 2: Train both agent. . **Crucially, disable the additive label reward by setting** $\lambda = 0$. This prevents the feedback agent from overfitting to imbalanced data (since the conversation agent generates fewer unsafe/overrefusing responses as it improves). The R_f^{DIR} is still conditioned on label correctness.

- Experiment Results
  - Dual Reduction in Safety and Rejection Rates: Compared to multiple baseline models, WALTZRL significantly lowers attack success rate (ASR)—for instance, reducing it from 39.0% to 4.6% on the WildJailbreak dataset—while simultaneously decreasing over-rejection rate (ORR) from 45.3% to 9.9%.
  - Preserved Capabilities: Even without utilizing generic “helpfulness” data during the RL phase, WALTZRL maintains the general capabilities and instruction-following abilities of the conversational agent, successfully pushing the Pareto front between safety and helpfulness.
  - Low latency impact: Thanks to an adaptive stopping mechanism, the feedback trigger rate (FTR) remains extremely low (only 6.7%) on non-safety-related general tasks (e.g., Alpaca Eval), ensuring efficient inference during real-world deployment.

- My Thought
  - Instead of being limited to unsafe-overrefuse, this positive-sum framework can work in other areas.
  - Coding: let the conversation agent be the coder and the feedback agent provide the review to code, such as time complexity, space, or some user preferences.
  - Complex Mathematical Problem: Let the conversation model form the CoT step by step, and let the feedback agent check the correctness of each step. If something goes wrong, then let the reviewer send a backtrack signal to make the conversation model return to the last correct step. (Similarly, tool-call)

- Work Cited
  - Paper
  - Chat History
  - NotebookLM